

DATA MINING

- **Data mining:** a process of filtering through large amounts of raw data to find useful information.
- **Artificial intelligence:** a computing tool that tries to operate in a way similar to the human brain.
- **Neural networks:** systems that continuously analyze and compare data until patterns emerge.
- **Clustering:** the process of dividing data into groups based on similar features or limited data ranges.
- **Decision trees:** models that separate data into subsets and analyze them across multiple levels.
- **Cleansed data:** data free from duplicate and erroneous information.
- **Data warehouse:** a storage system used to archive large amounts of data for easy access.
- **Trigger:** something that causes a process or action to start.
- **Discarded** = scrapping = removed
- **Made up** = invented
- **Mined** = extracted
- **Field** = category
- **Amounts** = quantities

Linux

- **GNU:** a free software project that provides tools and programs to create a free operating system.
- **GNOME:** a graphical desktop environment used on Linux systems.
- **Operating system:** software that manages computer hardware and software resources and allows users to run applications.
- **Directory:** a folder used to organize and store files on a computer.
- **Kernel:** the core of an operating system that handles memory allocation, communicates with hardware devices, and ensures the system runs properly.
- **Free Software Foundation:** an organization of volunteers dedicated to creating software that anyone can use, study, modify, and share for free.
- **Source code:** the original program code from which compiled programs are generated.
- **Open source:** a type of software development in which any programmer can study, modify, and fix bugs in the software.
- **A distribution:** a complete operating system package with the utilities and applications needed to make it useful.
- **X:** a standard distributed windowing system used to implement graphical interfaces on Unix/Linux systems.
- **KDE** : a graphical desktop environment for Linux.
- **Fulfilling** = satisfying

Acronym

- **KDE** : K Desktop Environment

Describing Technical Processes

- **VoIP**: a technology that allows voice communication over the Internet instead of traditional phone lines.
- **ATA (Analog Telephone Adapter)**: a device that connects a traditional telephone to a digital network (VoIP).
- **Dial**: to enter a phone number using a keypad to call someone.
- **Hooked up** = linked up
- **Switched on** = turned on
- **Boot up** = start up (a computer/system)
- **To perform** = to carry out

Acronym

- **VoIP** : Voice Over Internet Protocol

Active vs Passive — Rule

- **Active:**
Subject + Verb + Object
- **Passive:**
Object + be (conjugated) + Past Participle (+ by + agent)

Active / Passive Table (All Main Tenses)

Tense	Active Form	Passive Form	Active Example	Passive Example
Present Simple	S + base verb	O + am/is/are + PP	The programmer creates a website.	A website is created by the programmer.
Past Simple	S + past verb	O + was/were + PP	The programmer created a website.	A website was created by the programmer.
Future Simple	S + will + verb	O + will be + PP	The programmer will create a website.	A website will be created by the programmer.
Present Continuous	S + am/is/are + V-ing	O + am/is/are being + PP	The programmer is creating a website.	A website is being created by the programmer.

Past Continuous	S + was/were + V-ing	O + was/were being + PP	The programmer was creating a website.	A website was being created by the programmer.
Future Continuous	S+will be +V-ing	O +will be being + PP	The programmer will be creating a website.	A website will be being created by the programmer.
Present Perfect	S + have/has + PP	O + have/has been + PP	The programmer has created a website.	A website has been created by the programmer.
Past Perfect	S + had + PP	O + had been + PP	The programmer had created a website.	A website had been created by the programmer.
Future Perfect	S + will have + PP	O + will have been + PP	The programmer will have created a website.	A website will have been created by the programmer.
Present Perfect Continuous	S + have/has been + V-ing	O + have/has been being + PP	The programmer has been creating a website.	A website has been being created by the programmer.
Past Perfect Continuous	S + had been + V-ing	O + had been being + PP	The programmer had been creating a website.	A website had been being created by the programmer.
Future Perfect Continuous	S + will have been + V-ing	O + will have been being + PP	The programmer will have been creating a website.	A website will have been being created by the programmer.

Modal Verbs + Passive

Rule : **Modal + be + Past Participle**

Modal	Active	Passive
must	The programmer must create a website.	A website must be created by the programmer.
has to	The programmer has to create a website.	A website has to be created by the programmer.
should	The programmer should create a website.	A website should be created by the programmer.
can	The programmer can create a website.	A website can be created by the programmer.
ought to	The programmer ought to create a website.	A website ought to be created by the programmer.

Transformation Steps

1. **Object** → **Subject**
2. **Verb** → **be + past participle**
3. **Subject** → **by + agent (optional)**

Articles (a / an)

- **A** + singular countable noun
a pen ✓
a pens ✗
- **An** + singular noun starting with a vowel sound
an umbrella ✓
an umbrellas ✗

Sequencing a Process

Sequencing words are used to show the different stages of a process.

- **First ... then / next ... finally ...**
- **As** (while something is happening)
- **After / once**
- **Before**

Example:

First, the system is switched on. Then, it boots up. Finally, the user can make a call.

Safe Data Transfer

- **System**: a set of connected elements working together.
- **Cryptography**: the science of protecting information by encoding it so only authorized users can read it.
- **The code**: information that has been encoded to hide its meaning.
- **Protocol**: a set of rules that define how data is transmitted between systems.
- **Security**: protection of data against unauthorized access.
- **Privacy**: protection of personal or sensitive information.
- **Integrity**: assurance that data has not been altered or tampered with.
- **Encryption**: the process of converting readable data into an unreadable form.
- **Decryption**: the process of converting encrypted data back into readable form.
- **Authentication**: the process of verifying the identity of a user or system.
- **MAC (Message Authentication Code)**: a code used to verify data integrity and authenticity.
- **Gibberish**: meaningless or unreadable data.
- **Impostor**: a person pretending to be someone else.
- **Decipher**: to convert encoded data into meaningful data (déchiffrer).
- **Tamper**: to make unauthorized changes to data.
- **Tamper-proof**: protected against unauthorized changes (authentique).
- **Tenets**: principal features or fundamental principles.
- **The issuer**: the person or organization concerned / responsible (l'intéressé(e)).
- **Across** = via = along = through
- **Goal** = objective = purpose = aim
- **Prevent** = stop / avoid (empêcher)
- **Rely on** = based on
- **Encode** = encrypt
- **Decode** = decrypt
- **This** → these (plural)

Acronym

- **MAC** : Message Authentication Code

Crossword – Cyber World

- **SHIFT**: a keyboard key used to type capital letters or symbols
- **Caps Lock**: a key that allows typing letters in capital letters continuously.
- **Numeric keys**: keys used to enter numbers (0–9), usually on the keyboard or numeric keypad.

Acronyms

- **CPU**: Central Processing Unit
- **IBM**: International Business Machines
- **BIT**: Binary Digit
- **ROM**: Read Only Memory
- **RAM**: Random Access Memory

Crossword – Computer Science

- **Bug**: an error or flaw in a computer program that causes it to work incorrectly.
- **Clock speed**: the rate at which a computer processor executes instructions, usually measured in hertz (Hz).
- **Virus**: a malicious program that can copy itself and damage or disrupt a computer system.
- **Plagiarism**: the act of copying someone else's work or ideas and presenting them as your own.

Crossword – Cyber Security

- **Social engineering**: the use of psychological manipulation to trick people into revealing confidential information.
- **Password manager**: a tool that securely stores and manages passwords.
- **Firewall**: a security system that monitors and controls incoming and outgoing network traffic.
- **Cyber hygiene**: good practices that help maintain security and protect systems and data.
- **Tailgating**: an attack where an unauthorized person gains physical access by following an authorized person.
- **Intranet**: a private network accessible only to members of an organization.
- **Phishing**: a cyberattack that uses fake emails or messages to steal personal or sensitive information.
- **Ransomware**: malicious software that locks or encrypts data and demands payment to restore access.

Cours où il y a 3 textes (Programming Expert, Computer Consultant et IT Manager)

- **Hop** = *saut*
- **Leap** = *bond* (big jump)
- **Junior / Senior** = beginner / experienced
- **Stuff** = things / equipment

- **In-house** = internal (within the company)
- **Rolled out** = launched / introduced
- **Tailored** = customized / adapted
- **In-house** = internal
- **Interestingly** = in an interesting or surprising way

Crossword – Artificial Intelligence (AI)

- **Check in shelves**: to organize or monitor items on shelves (vérifier les étagères / inventorier)
- **Healthcare**: the system or practice of maintaining health (soins de santé, secteur médical)
- **Herbicides**: chemicals used to kill unwanted plants (herbicides, produits chimiques pour plantes)
- **Underway**=in progress, happening (en cours, en train de se faire)

Crossword –Robotics And Artificial Intelligence

- **Multipurpose** : designed to serve many functions
- **Neural Network (AI)** : a system of algorithms modeled after the human brain to recognize patterns and learn
- **Synoid** = humanoid robot, often female in appearance (female robot, humanoid robot)
- **Cyborg** = part-human, part-machine (human-robot hybrid)
- **Purpose** = goal, objective, aim

Acronym

- **FBI** :Federal Bureau of Investigation